

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 38)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

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0 1 . 2

Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

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0 1 . 3

Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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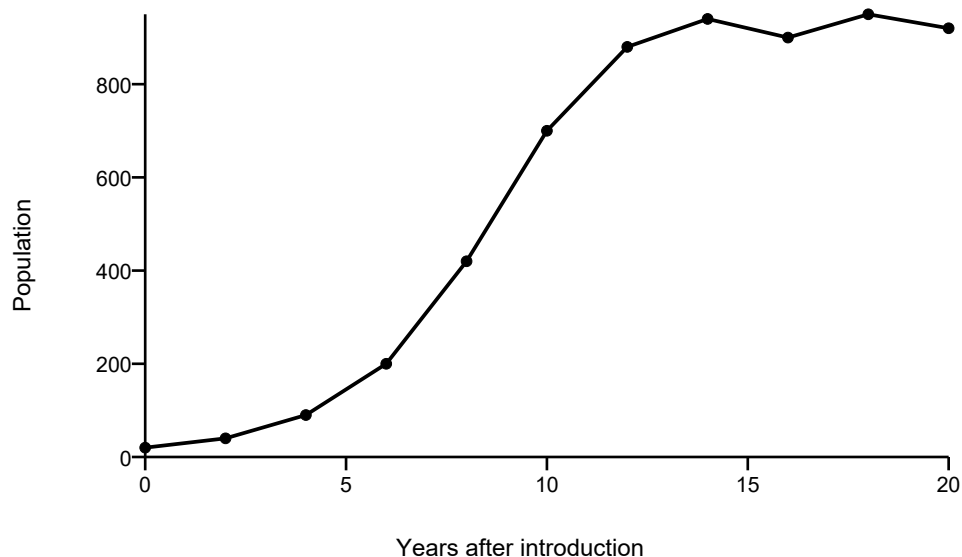
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10

0 2

Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 2 . 1

Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

0 2 . 2

Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

0 2 . 3

Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 2 . 4

Explain how predator and prey populations regulate each other.

[2 marks]

10

0 3 . 1

Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 3 . 2

Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

03 . 3

Explain how buying fairtrade food affects producers.

[1 mark]

03 . 4

Explain how nitrate from farmland can threaten human health.

[2 marks]

03 . 5

Give **two** agricultural sources of methane.

[2 marks]

1

2

10

04

Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0	4	.	1
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Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

0	4	.	2
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Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

0	4	.	3
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Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

0 5 Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

0 5 . **1** Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

0 5 . **2** Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 5 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

0 6

Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	1	2	4	6	8	10	12	14
Running mean catch per night	31	18	24	21	22	22	21	22

0 6 . 1

Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

0 6 . 2

Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

06 . 3

Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

06 . 4

Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

10

07 . 1

State what is meant by **hybrid vigour**.

[1 mark]

07 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

07 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

08 . 1

For each of the following forms of direct exploitation, give a named species whose population has been reduced.

[5 marks]

Food

.....

Fashion

.....

Entertainment / pets

.....

Furniture and ornaments

.....

Traditional medicines

.....

08 . 2 Explain why species are deliberately eradicated. Give **three** categories with examples.

[3 marks]

1

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2

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3

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08 . 3 Explain why legal protection alone rarely prevents the decline of a species exploited for traditional medicine.

[2 marks]

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10

0	9	.	1
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Explain what a carbon footprint should include when calculated for a car.

[3 marks]

0	9	.	2
---	---	---	---

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

0	9	.	3
---	---	---	---

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

10

1	0	.	1
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Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

Extra space

1	0
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2

 State what is meant by **biodiversity**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

1 1 . 2

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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Do not write outside the box

END OF QUESTIONS